Atharva Chandak

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WORK EXPERIENCE

Ebay **SEPT 2025 – PRESENT**

ML Engineer 2 | Python, Pytorch, LangChain *Bengaluru, India*

- Building **Multi-Modal model** for matching user listing items to catalog products for over **1 Billion+** listings on the platform.
- Researching and **finetuning SOTA VLLMs** for **fine-grained image data extraction** in Sports Trading Cards segment.
- Prototyped a buyer experience assistant for **hyper-personalised** recommendations and **agentic assistance** to users using memory.

Wells Fargo **JUL 2023 – SEPT 2025**

SWE | Python, Java, JavaScript, React, Spring, Jenkins, MySQL *Bengaluru, India*

- **Full-stack developer** for high-volume PL credit card and loan platforms that processed **1M+** transactions monthly for 18k+ merchants.
- Prototyped a secure & scalable **LLM based chatbot for merchants** to help simplify UX and reducing loan application time by **30%**.
- Developed backend **API micro-services** for POS EMIs increasing transaction throughput and ensuring frictionless payments & processing.
- Built a SaaS-based **customer insights portal** aggregating merchant data, boosting segmentation & targeting accuracy for data-driven decisions.

Mila / Robotics and Embodied AI Lab, University of Montreal **JAN 2023 – JUN 2023**

Research Assistant | Python, Pytorch, Hugging Face *Montreal, Canada*

- Researched **Continual Object Detection** in autonomous driving for my undergrad thesis, cutting catastrophic forgetting in OD by **73%**.
- Created on-device incremental OD methods for robotics demonstrating **mAP50 drop** of only **3.99** for transformer models vs CNNs' 14.84.
- Achieved a domain incremental **avg mAP50 of 68.63** and **avg mAP of 43.89** on the CLAD-D dataset marking **15% gain** over baselines.

Wells Fargo **MAY 2022 – JUL 2022**

SDE Intern | JavaScript, Python, React, Flask, MongoDB *Hyderabad, India (Remote)*

- Designed and implemented an **AI-powered voice chatbot** for home lending, reducing query resolution time by **up to 68%**.
- Integrated **speech-to-text**(STT) and **TTS** APIs for voice interaction, and built an action server for NLP-driven processing & response curation.
- Composed the **NLP pipeline** for intent and entity extraction and achieved an **F1 scores** of **93.4%** and **83.5%**.

Airlab, Carnegie Mellon University **AUG 2021 – DEC 2021**

Intern | Python, Pytorch | Paper Link *Pittsburgh, USA (Remote)*

- Conceived **the vision pipeline** for TartanAviation, generating **3.1M** high-resolution images across 550 terminal airspace sequences.
- Engineered calibration and **auto-labeling frameworks** for precise aircraft detection, producing **700k+** high-fidelity bounding box annotations.

EDUCATION

Birla Institute of Technology and Science Pilani (BITS Pilani)

Bachelor of Engineering in Computer Science

Coursework : Data Structures & Algorithms (DSA), Object Oriented Programming (OOP) design patterns, Database Systems, Computer Networks, Cloud Computing, Operating Systems, Probability & Statistics, Machine Learning, Deep Learning, Computer Vision (CV), Natural Language Processing

PUBLICATIONS AND PROJECTS

AdQuestA: Knowledge-Guided Visual Question Answer Framework For Advertisements | Paper Link **WACV (FEB 2025)**

- Proposed a novel VQA framework for advertisements achieving **pattern-matching** accuracy of **58.58%**, outperforming 10X larger models.
- Introduced **first** ad-specific **VQA dataset** featuring **94k+ QA pairs** & **17k images**, establishing a new benchmark for ad comprehension research.
- Integrated **RAG** and **LLaVA**-based implicit knowledge, leading to **2X increase** in QA accuracy compared to previous state-of-the-art approaches.

LIFTNet: Action Recognition Across Varied Lighting Environments **MAY 2022 – DEC 2022**

- Architected LIFT-Net, a **dual-pathway transformer**, exceeding SOTA by **2.20%** (HMDB51) and **0.12%** (UCF101) for action recognition.
- Enhanced **illumination robustness** by efficient spatiotemporal modeling, achieving **95.21%** on ARID (+**1.14%**) and **2.65%** gain on InfAR.

Drone for Parcel Delivery **OCT 2021 – DEC 2021**

- Optimized drone delivery and returns **increasing deliveries** by **23%** and **reducing errors** by **31%** within mission constraints.
- Executed control systems, path planning, and QR-based detection algorithms - boosting **UAV navigation accuracy** by **27%** over baselines.

TECHNICAL SKILLS

Programming Languages : Python, JavaScript, TypeScript, HTML/CSS, C++
Machine Learning Libraries : Pytorch, Hugging Face, LangChain, Numpy, pandas, OpenCV, scikit-learn, Tensorflow, Pytorch-lightning, FastAI
Development Frameworks : Nodejs, NextJS, React, Redux, Spring, Flask, Django, JUnit, Mockito, Jest, REST API, Kafka, MySQL, MongoDB
Tools/Technologies : Docker, Linux/Unix, AWS & Azure Cloud, Git, Agile, SDLC, Jenkins (CI/CD), ROS

LEADERSHIP / VOLUNTEERING

BITS-ACM Core Team Managed a software development hackathon (HackBITSPilani); Contributed and managed org's open source projects.
Senior Member, ShARE BITS Pilani Interned at Catalytic Corps, developing a framework to analyze and solve Indian MSME sector challenges.
Teaching Assistant Orchestrated ML/DL course assignments and conducted workshops on programming neural nets and foundational AI concepts.
NSS Volunteer Taught computer literacy and organized social impact events like blood donation camps, youth conferences, etc. in Pilani.

MAJOR ACHIEVEMENTS

Secured 3rd among 148 teams in AI RoboSoccer Competition; Trained a PPO-based RL agent maximizing goal scoring and passing accuracy.
Ranked 598 & 2449 out of 1.23 million students in JEE MAIN & JEE Advanced examination.
Top 5 among 472 teams, in E-Yantra Robotics innovation Competition hosted by IIT Bombay for simulated autonomous drone algorithms.